JAGADISH JANAKIRAMAN

Performance Engineering Lead | Banking & Healthcare

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PROFESSIONAL SUMMARY

Performance Engineering Lead with 20+ years across banking and healthcare. Defines SLAs/SLOs and workload models, leads load, spike, chaos and failover testing on Azure/AKS, and turns results into executive go/no-go decisions. Scaled Kaiser Permanente's pharmacy platform 10x across 400+ stores, managed a 30-member team at Accenture, and led performance engineering for TD Bank's branch account-opening platform migration and its document verification services. Builds open-source performance tooling for scripting, load execution, service virtualisation, chaos and regression analysis.

CORE COMPETENCIES

Program Strategy: Enterprise performance framework design, SLA/KPI/TPS definition, workload modelling, performance risk register, governance across multi-vendor programs

Load Testing: JMeter, LoadRunner Enterprise & Cloud — Web, Mobile, API and banking workloads

Web & UI: Selenium (incl. an in-house JavaScript framework), Puppeteer, Chrome mobile emulation

API & Middleware: REST/HTTP, SOAP, Postman, MuleSoft API layer testing, service virtualisation — CA DevTest, Ready API, WireMock

APM & Observability: Dynatrace (11 years), Splunk, Datadog, Grafana, InfluxDB

Cloud & Container: Azure (AKS, App Services, Monitor), Docker, Kubernetes — HPA tuning, pod scaling, resource limits

CI/CD & DevOps: GitHub Actions (on-demand JMeter suites with threshold-based pass/fail), Jenkins (CI-integrated performance regression)

Test Strategy: NFR definition, p95/p99 profiling, breakpoint, endurance, soak, spike, chaos (Chaos Monkey), failover, alert validation

Databases: Oracle, PostgreSQL, Cosmos DB — query plan analysis & tuning

Languages: Java (Spring Boot stub services), JavaScript / Node.js, Python, Shell Scripting, SQL

AI-Assisted Engineering: GitHub Copilot (test scripts, test data, y4m camera-feed video generation), Ollama (local LLM analysis of results)

Leadership: PMP, SAFe, Scrum — executive reporting (Power BI, Tableau), cross-functional teams, vendor management, mentoring

PROFESSIONAL EXPERIENCE

Performance Engineering Lead · TD Bank (Toronto-Dominion Bank)

Toronto, ON | November 2022 – Present

- Single point of accountability for performance engineering (2022–2024) on TD's branch platform used by customer service reps to open accounts, sell products and update customer details, through its migration to a new digital platform. Since 2024, leading performance testing for the bank's document verification services.
- Defined program-wide SLAs, KPIs, TPS targets and latency thresholds for customer-facing Web and API endpoints; established the delivery process from intake through workload modelling, test data, environment readiness, defect triage and executive confidence reporting.
- Pioneered TD's first performance testing for document verification by injecting live camera-feed input (document and selfie capture) with y4m video at runtime — no existing tooling could do it; later cut test-video generation from hours to minutes with a GitHub Copilot agent.
- Resolved document verification bottlenecks: traced 20–30% stub-server call failures to TCP handshake failures (fixed with network and Linux teams), found an undersized HTTP connection pool in a BFF microservice through Dynatrace, and tuned slow audit-database queries (indexes, statistics, data-type fixes).
- Identified a critical HPA misconfiguration through spike testing after production showed surges to ~2,000 TPS at the top of every hour against a 700 TPS tested peak; lowered the scaling threshold with the infrastructure team and validated corrected autoscaling.
- Right-sized ~40 Azure App Service apps on the branch platform after utilization and capacity testing, saving about CAD 50,000 per month.
- Introduced chaos testing on AKS with Chaos Monkey: a 2-second injected downstream delay produced 20–30-second responses; traced connection-pool exhaustion in Dynatrace and fixed it by resizing the pool. Also run failover and alert-validation tests.
- Built UI performance scripts on the bank's in-house JavaScript/Selenium framework, using GitHub Copilot to generate scripts from recorded Puppeteer flows instead of hand-writing XPaths; built GitHub Actions workflows for on-demand JMeter suites with automated response-time and error-rate thresholds.
- Defined SLI/SLO targets and built Dynatrace and Splunk dashboards for p95/p99 latency, throughput and incident triage — giving release teams go/no-go signals and executives readiness scorecards.

Performance Engineering Lead / Manager · Kaiser Permanente (via DXC Technology, then Accenture)
San Francisco, CA | August 2011 – October 2022

- Led performance engineering for Kaiser's Pharmacy Modernisation program across multiple vendor teams, agile workstreams and enterprise architecture stakeholders; validated on-premises-to-Azure and IBM Bluemix migrations using production traces as load models.
- Scaled the Kaiser Pharmacy platform 10x across 400+ stores within 24 months; tuned systems to sustain 3x peak demand during COVID-19 by running breakpoint and soak tests before demand surges hit production.
- Designed a Video Conference Performance Testing Framework using LoadRunner, Selenium, Java and Power BI dashboards — executive-ready coverage of end-to-end latency, error rate and throughput for 400+ stores.
- Integrated JMeter and LoadRunner into Jenkins CI/CD pipelines, embedding non-functional regression testing into sprint releases and catching performance defects earlier.
- Used Dynatrace distributed tracing across pharmacy microservices for method-level hotspot analysis and fast triage of performance issues.
- Managed a 30-member performance engineering team in India at Accenture alongside onshore US teams; mentored test leads and set scripting standards adopted across vendor teams.

Performance Test Engineer · Tata Consultancy Services (TCS)
India & USA | April 2004 – August 2011

- Delivered performance testing for Bank of America, Citibank and Deluxe Corporation, resolving throughput and latency bottlenecks on high-volume banking applications.

PROJECTS

- **Perf Script Assistant** (github.com/jagsrj/perf-script-assistant) — records browser workflows in Chrome/Edge and generates correlated JMeter and Selenium scripts; rule-based auto-correlation, with optional Claude review of uncertain values.
- **JMX Lint** (github.com/jagsrj/jmeter-script-linter) — JMeter plugin and CI quality gate for test plans; 45 rules across naming, memory risk, best practice and security, with JUnit/JSON reports for GitHub Actions and Jenkins.
- **Performance Testing Framework** (github.com/jagsrj/perf-framework) — self-hosted JMeter console (Node.js, Docker, InfluxDB, Grafana); distributed load generators, live SLO aborts, automatic baselines, CI gate and an MCP server.
- **Service Virtualizer** (github.com/jagsrj/service-virtualizer) — WireMock-based virtual services with p50/p95 latency profiles, error injection and thread-pool capacity limits, switchable mid-test through a control API.
- **Chaos Runner** (github.com/jagsrj/chaos-runner) — hypothesis-driven chaos experiments on Kubernetes/AKS with steady-state checks, crash-safe rollback, recovery-time measurement and production guardrails.
- **Perf Insight** (github.com/jagsrj/perf-insight) — results analyzer and CI regression gate; statistical comparison against a rolling baseline (Mann-Whitney U), SLO checks and Dynatrace/Splunk correlation.

EDUCATION

Master of Science, Data Science — University of Arizona, USA
Bachelor of Electronics & Communications Engineering — University of Madras, India

CERTIFICATIONS

PMP · Microsoft Azure Fundamentals · Certified SAFe Agilist · Certified Scrum Professional · Dynatrace Associate · Splunk User · ITIL Foundation · HP Certified LoadRunner Specialist · Sun Certified Java Programmer